PROGRAMMING IN JAVA

CSA0915

ASSIGNMENT-CO1

CAMPUS SMART-PARKING SYSTEM

TASK 10 – Abstraction and Interfaces

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1. Task10.md

Aim

To demonstrate encapsulation, packages and access modifiers in Java using the Campus Smart-Parking System.

Encapsulation

Encapsulation is the process of wrapping data and methods into a single class and protecting data from direct access.

Access Modifiers

1. **Private:** Accessible only within the same class.
2. **Default:** Accessible within the same package.
3. **Protected:** Accessible within the same package and subclasses.
4. **Public:** Accessible from anywhere.

Packages

Packages are used to organize related Java classes and avoid naming conflicts.

Application

The ParkingAccount class stores vehicle owner details and parking balance using private variables. Public getter and setter methods provide controlled access.

Advantages

- Data security
- Code organization

- Easy maintenance
- Controlled access to data

Conclusion

Encapsulation protects parking account information, while packages and access modifiers improve code organization and security.

2. ParkingAccount.java

```
package parking;

public class ParkingAccount {

    // Private data members

    private String ownerName;

    private double balance;

    // Default access modifier

    String accountType = "Student";

    // Constructor

    public ParkingAccount(String ownerName, double balance) {

        this.ownerName = ownerName;

        this.balance = balance;

    }

    // Public getter

    public String getOwnerName() {

        return ownerName;

    }

    // Public setter

    public void setOwnerName(String ownerName) {
```

```
        this.ownerName = ownerName;
    }
    public double getBalance() {
        return balance;
    }
    // Protected method
    protected void displayStatus() {
        System.out.println("Account Status: Active");
    }
    // Public method to add balance
    public void addBalance(double amount) {
        if (amount > 0) {
            balance += amount;
        }
    }
    public void displayDetails() {
        System.out.println("Owner Name: " + ownerName);
        System.out.println("Account Type: " + accountType);
        System.out.println("Balance: Rs. " + balance);
        displayStatus();
    }
}
```

3. Main.java

```
package parking;

public class Main {

    public static void main(String[] args) {

        ParkingAccount account =

            new ParkingAccount("Sreya", 500);

        System.out.println("Campus Parking Account Details:");

        account.displayDetails();

        account.addBalance(200);

        System.out.println();

        System.out.println("After Adding Balance:");

        account.displayDetails();

        account.setOwnerName("Reddy Sreya");

        System.out.println();

        System.out.println("Updated Owner: " +

                        account.getOwnerName());

    }

}
```

4. Output:

